School Management System

REST API Design — Enterprise Edition

`Base URL: http://localhost:8080`

Architect: Senior Java Enterprise Application Architect

API Overview

This document defines all business-usable REST endpoints for the School Management System. Every endpoint follows RESTful conventions with proper HTTP methods, meaningful URIs, and structured JSON request/response bodies.

Domain	Endpoints	Description
Students	5	CRUD operations on student records
Subjects	5	CRUD operations on subject catalog
Exams	5	CRUD operations on exam events
Marks	5	CRUD operations on score records
Performance Analytics	5	Student performance reports & trends
Reports & Leaderboards	4	Top performers, pass/fail, subject averages

1. Students API

1.1 Get All Students

GET	/api/students	Retrieve a list of all registered students

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "studentId": 1,
      "firstName": "Aarav",
      "lastName": "Sharma",
      "enrollmentYear": 2023,
      "fullName": "Aarav Sharma"
    },
    {
      "studentId": 2,
      "firstName": "Diya",
      "lastName": "Patel",
      "enrollmentYear": 2023,
      "fullName": "Diya Patel"
    }
  ],
  "totalCount": 4
}
```

1.2 Get Student by ID

GET	/api/students/{studentId}	Retrieve a single student by their unique ID

Path Parameters:

Parameter	Type	Description
studentId	`Integer`	Unique identifier of the student

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "studentId": 1,
    "firstName": "Aarav",
    "lastName": "Sharma",
    "enrollmentYear": 2023,
    "fullName": "Aarav Sharma"
  }
}
```

Response — 404 Not Found

```
{
  "status": "error",
  "errorCode": "STUDENT_NOT_FOUND",
  "message": "Student with ID 99 does not exist."
}
```

1.3 Create a New Student

POST	/api/students	Register a new student in the system

Request Body

```
{
  "firstName": "Rohan",
  "lastName": "Mehta",
  "enrollmentYear": 2025
}
```

Response — 201 Created

```
{
  "status": "success",
  "message": "Student created successfully.",
  "data": {
    "studentId": 5,
    "firstName": "Rohan",
    "lastName": "Mehta",
    "enrollmentYear": 2025,
    "fullName": "Rohan Mehta"
  }
}
```

1.4 Update a Student

PUT	/api/students/{studentId}	Update personal details of an existing student

Path Parameters:

Parameter	Type	Description
studentId	`Integer`	Unique identifier of the student to update

Request Body

```
{
  "firstName": "Rohan",
  "lastName": "Kumar",
  "enrollmentYear": 2025
}
```

Response — 200 OK

```
{
  "status": "success",
  "message": "Student updated successfully.",
  "data": {
    "studentId": 5,
    "firstName": "Rohan",
    "lastName": "Kumar",
    "enrollmentYear": 2025,
    "fullName": "Rohan Kumar"
  }
}
```

1.5 Delete a Student

DELETE	/api/students/{studentId}	Permanently delete a student record

Path Parameters:

Parameter	Type	Description
studentId	`Integer`	Unique identifier of the student to delete

Response — 200 OK

```
{
  "status": "success",
  "message": "Student with ID 5 has been deleted successfully."
}
```

HTTP Response Codes:

HTTP Status	Meaning
`200 OK`	Student deleted successfully
`404 Not Found`	Student with given ID not found
`409 Conflict`	Cannot delete — student has existing marks records

2. Subjects API

2.1 Get All Subjects

GET	/api/subjects	Retrieve the full list of subjects offered

Response — 200 OK

```
{
  "status": "success",
  "data": [
    { "subjectId": 101, "subjectName": "Mathematics" },
    { "subjectId": 102, "subjectName": "Science" },
    { "subjectId": 103, "subjectName": "Hindi Literature" }
  ],
  "totalCount": 3
}
```

2.2 Get Subject by ID

GET	/api/subjects/{subjectId}	Retrieve a single subject by ID

Path Parameters:

Parameter	Type	Description
subjectId	Integer	Unique identifier of the subject

Response — 200 OK

```
{
  "status": "success",
  "data": { "subjectId": 101, "subjectName": "Mathematics" }
}
```

2.3 Create a New Subject

POST	/api/subjects	Add a new subject to the curriculum

Request Body

```
{
  "subjectName": "English Literature"
}
```

Response — 201 Created

```
{
  "status": "success",
  "message": "Subject created successfully.",
  "data": { "subjectId": 104, "subjectName": "English Literature" }
}
```

2.4 Update a Subject

PUT	/api/subjects/{subjectId}	Rename or update an existing subject

Path Parameters:

Parameter	Type	Description
subjectId	`Integer`	Unique identifier of the subject to update

Request Body

```
{ "subjectName": "Advanced Mathematics" }
```

Response — 200 OK

```
{
  "status": "success",
  "message": "Subject updated successfully.",
  "data": { "subjectId": 101, "subjectName": "Advanced Mathematics" }
}
```

2.5 Delete a Subject

DELETE	`/api/subjects/{subjectId}`	Remove a subject from the system

Path Parameters:

Parameter	Type	Description
subjectId	`Integer`	Unique identifier of the subject to delete

Response — 200 OK

```
{
  "status": "success",
  "message": "Subject with ID 104 has been deleted successfully."
}
```

3. Exams API

3.1 Get All Exams

GET	/api/exams	Retrieve all exam events (supports ?sort=date)

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "examId": 1,
      "examName": "Half-Yearly 2024",
      "examDate": "2024-09-15"
    },
    {
      "examId": 2,
      "examName": "Final Annual 2024",
      "examDate": "2024-03-20"
    }
  ],
  "totalCount": 2
}
```

3.2 Get Exam by ID

GET	/api/exams/{examId}	Retrieve a specific exam event by its ID

Path Parameters:

Parameter	Type	Description
examId	`Integer`	Unique identifier of the exam event

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "examId": 1,
    "examName": "Half-Yearly 2024",
    "examDate": "2024-09-15"
  }
}
```

3.3 Create a New Exam

POST	/api/exams	Schedule a new exam event

Request Body

```
{
  "examName": "Unit Test 1 - 2025",
  "examDate": "2025-07-10"
}
```

Response — 201 Created

```
{
  "status": "success",
  "message": "Exam created successfully.",
  "data": {
    "examId": 3,
    "examName": "Unit Test 1 - 2025",
    "examDate": "2025-07-10"
  }
}
```

3.4 Update an Exam

PUT	/api/exams/{examId}	Reschedule or rename an existing exam

Path Parameters:

Parameter	Type	Description
examId	Integer	Unique identifier of the exam to update

Request Body

```
{
  "examName": "Unit Test 1 - 2025 (Revised)",
  "examDate": "2025-07-15"
}
```

Response — 200 OK

```
{
  "status": "success",
  "message": "Exam updated successfully.",
  "data": {
    "examId": 3,
    "examName": "Unit Test 1 - 2025 (Revised)",
    "examDate": "2025-07-15"
  }
}
```

3.5 Delete an Exam

DELETE	/api/exams/{examId}	Remove an exam event record

Path Parameters:

Parameter	Type	Description
examId	Integer	Unique identifier of the exam to delete

Response — 200 OK

```
{
  "status": "success",
  "message": "Exam with ID 3 has been deleted successfully."
}
```

HTTP Response Codes:

HTTP Status	Meaning

`200 OK`	Exam deleted successfully
`404 Not Found`	Exam with given ID not found
`409 Conflict`	Cannot delete — exam has associated marks records

4. Marks API

4.1 Get All Marks (with Filters)

GET	/api/marks	Retrieve marks with optional query filters

Query Parameters (all optional):

Parameter	Type	Description
studentId	Integer	Filter marks by a specific student
subjectId	Integer	Filter marks by a specific subject
examId	Integer	Filter marks by a specific exam
minScore	Integer	Return only marks above this score threshold

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "markId": 1,
      "student": { "studentId": 1, "fullName": "Aarav Sharma" },
      "subject": { "subjectId": 101, "subjectName": "Mathematics" },
      "exam":    { "examId": 1, "examName": "Half-Yearly 2024" },
      "score": 85
    },
    {
      "markId": 2,
      "student": { "studentId": 1, "fullName": "Aarav Sharma" },
      "subject": { "subjectId": 102, "subjectName": "Science" },
      "exam":    { "examId": 1, "examName": "Half-Yearly 2024" },
      "score": 90
    }
  ],
  "totalCount": 8
}
```

4.2 Get Mark by ID

GET	/api/marks/{markId}	Retrieve a specific mark entry by its ID

Path Parameters:

Parameter	Type	Description
markId	Integer	Unique identifier of the mark record

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "markId": 1,
    "student": { "studentId": 1, "fullName": "Aarav Sharma" },
```

```
    "subject": { "subjectId": 101, "subjectName": "Mathematics" },
    "exam":    { "examId": 1, "examName": "Half-Yearly 2024" },
    "score": 85
  }
}
```

4.3 Record a New Mark

POST	/api/marks	Enter a score for a student in a specific subject and exam

Request Body

```
{
  "studentId": 4,
  "subjectId": 102,
  "examId": 1,
  "score": 78
}
```

Response — 201 Created

```
{
  "status": "success",
  "message": "Mark recorded successfully.",
  "data": {
    "markId": 9,
    "student": { "studentId": 4, "fullName": "Ananya Singh" },
    "subject": { "subjectId": 102, "subjectName": "Science" },
    "exam":    { "examId": 1, "examName": "Half-Yearly 2024" },
    "score": 78
  }
}
```

Response — 409 Conflict

```
{
  "status": "error",
  "errorCode": "DUPLICATE_MARK",
  "message": "A mark already exists for Student 4, Subject 102, Exam 1."
}
```

4.4 Update a Mark (Correction)

PUT	/api/marks/{markId}	Correct or update a previously recorded score

Path Parameters:

Parameter	Type	Description
markId	`Integer`	Unique identifier of the mark to update

Request Body

```
{ "score": 82 }
```

Response — 200 OK

```
{
  "status": "success",
  "message": "Mark updated successfully.",
  "data": {
    "markId": 9,
```

```
    "student": { "studentId": 4, "fullName": "Ananya Singh" },
    "subject": { "subjectId": 102, "subjectName": "Science" },
    "exam":    { "examId": 1, "examName": "Half-Yearly 2024" },
    "score": 82
  }
}
```

4.5 Delete a Mark

DELETE	`/api/marks/{markId}`	Remove a mark entry from the system

Path Parameters:

Parameter	Type	Description
markId	`Integer`	Unique identifier of the mark to delete

Response — 200 OK

```
{
  "status": "success",
  "message": "Mark with ID 9 has been deleted successfully."
}
```

5. Performance Analytics API

5.1 Get Full Performance Report for a Student

GET	/api/performance/students/{studentId}	Detailed performance of a student across all subjects and exams

Path Parameters:

Parameter	Type	Description
studentId	Integer	The student whose performance to retrieve

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "student": { "studentId": 1, "fullName": "Aarav Sharma", "enrollmentYear": 2023 },
    "overallAverage": 87.67,
    "totalExamsAppeared": 2,
    "performance": [
      {
        "examId": 1,
        "examName": "Half-Yearly 2024",
        "examDate": "2024-09-15",
        "subjects": [
          { "subjectName": "Mathematics", "score": 85, "status": "Pass" },
          { "subjectName": "Science",     "score": 90, "status": "Pass" }
        ],
        "examTotal": 175,
        "examAverage": 87.5
      },
      {
        "examId": 2,
        "examName": "Final Annual 2024",
        "examDate": "2024-03-20",
        "subjects": [
          { "subjectName": "Mathematics", "score": 88, "status": "Pass" }
        ],
        "examTotal": 88,
        "examAverage": 88.0
      }
    ]
  }
}
```

5.2 Get Student Performance in a Specific Exam

GET	/api/performance/students/{studentId}/exams/{examId}	All subject scores for one student in one exam

Path Parameters:

Parameter	Type	Description
studentId	Integer	Target student ID

examId	`Integer`	Target exam ID

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "student": { "studentId": 1, "fullName": "Aarav Sharma" },
    "exam": { "examId": 1, "examName": "Half-Yearly 2024", "examDate": "2024-09-15" },
    "scores": [
      { "subjectName": "Mathematics", "score": 85, "status": "Pass" },
      { "subjectName": "Science",     "score": 90, "status": "Pass" }
    ],
    "totalScore": 175,
    "averageScore": 87.5
  }
}
```

5.3 Get Average Scores per Subject (All Exams)

GET	**`/api/performance/subjects/averages`**	Calculate average score for every subject across all exams

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "subjectId": 101,
      "subjectName": "Mathematics",
      "averageScore": 85.83,
      "totalMarksRecorded": 6,
      "highestScore": 95,
      "lowestScore": 75
    },
    {
      "subjectId": 102,
      "subjectName": "Science",
      "averageScore": 75.0,
      "totalMarksRecorded": 2,
      "highestScore": 90,
      "lowestScore": 60
    }
  ]
}
```

5.4 Get Top Performers (Global)

GET	**`/api/performance/top-performers`**	Returns the student with the highest single score across all exams

Query Parameters (optional):

Parameter	Type	Description
limit	`Integer`	Number of top performers to return (default: 5)

examId	`Integer`	Scope top performers to a specific exam

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "rank": 1,
      "student": { "studentId": 3, "fullName": "Vihaan Rao" },
      "exam": { "examName": "Half-Yearly 2024" },
      "subject": { "subjectName": "Mathematics" },
      "highestScore": 95
    },
    {
      "rank": 2,
      "student": { "studentId": 1, "fullName": "Aarav Sharma" },
      "exam": { "examName": "Half-Yearly 2024" },
      "subject": { "subjectName": "Science" },
      "highestScore": 90
    }
  ]
}
```

5.5 Pass / Fail Report for an Exam

GET	**`/api/performance/exams/{examId}/pass-fail`**	Pass/Fail status for all students in a given exam (pass threshold: 70)

Path Parameters:

Parameter	Type	Description
examId	`Integer`	The exam for which to generate the pass/fail report

Query Parameters (optional):

Parameter	Type	Description
passThreshold	`Integer`	Minimum score to pass (default: 70)

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "exam": { "examId": 1, "examName": "Half-Yearly 2024", "examDate": "2024-09-15" },
    "passThreshold": 70,
    "summary": { "totalStudents": 3, "passed": 2, "failed": 1, "passPercentage": 66.67 },
    "results": [
      {
        "student": { "studentId": 1, "fullName": "Aarav Sharma" },
        "subjectResults": [
          { "subjectName": "Mathematics", "score": 85, "status": "Pass" },
          { "subjectName": "Science",     "score": 90, "status": "Pass" }
        ],
        "overallStatus": "Pass"
      },
      {
        "student": { "studentId": 2, "fullName": "Diya Patel" },
        "subjectResults": [
```

```
          { "subjectName": "Mathematics", "score": 75, "status": "Pass" },
          { "subjectName": "Science",     "score": 60, "status": "Fail" }
        ],
        "overallStatus": "Fail"
      }
    ]
  }
}
```

6. Reports & Leaderboards API

6.1 Exam Leaderboard (Total Score Ranking)

GET	`/api/reports/exams/{examId}/leaderboard`	Rank all students by their total score in a specific exam

Path Parameters:

Parameter	Type	Description
examId	`Integer`	The exam whose leaderboard to retrieve

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "exam": { "examId": 1, "examName": "Half-Yearly 2024" },
    "leaderboard": [
      { "rank": 1, "student": { "studentId": 3, "fullName": "Vihaan Rao" },
"totalScore": 95,  "subjectsAttempted": 1 },
      { "rank": 2, "student": { "studentId": 1, "fullName": "Aarav Sharma" },
"totalScore": 175, "subjectsAttempted": 2 },
      { "rank": 3, "student": { "studentId": 2, "fullName": "Diya Patel" },
"totalScore": 135, "subjectsAttempted": 2 }
    ]
  }
}
```

6.2 Exam Total Score Summary (Exams above a Threshold)

GET	`/api/reports/exams/summary`	Returns exams where total score sum exceeds a given threshold

Query Parameters:

Parameter	Type	Description
minTotalScore	`Integer`	Only return exams where sum of scores exceeds this value (default: 0)

Response — 200 OK

```
{
  "status": "success",
  "data": [
    {
      "examId": 1,
      "examName": "Half-Yearly 2024",
      "examDate": "2024-09-15",
      "totalScoreSum": 405,
      "studentsAppeared": 3
    },
    {
      "examId": 2,
      "examName": "Final Annual 2024",
      "examDate": "2024-03-20",
      "totalScoreSum": 259,
```

```
      "studentsAppeared": 3
    }
  ]
}
```

6.3 Compare Two Students Head-to-Head

GET	`/api/reports/students/compare`	Side-by-side performance comparison between two students

Query Parameters (required):

Parameter	Type	Description
studentIdA	`Integer`	First student ID for comparison
studentIdB	`Integer`	Second student ID for comparison
examId	`Integer`	(Optional) Limit comparison to a specific exam

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "studentA": {
      "student": { "studentId": 1, "fullName": "Aarav Sharma" },
      "overallAverage": 87.67,
      "subjectAverages": [
        { "subjectName": "Mathematics", "average": 86.5 },
        { "subjectName": "Science",     "average": 90.0 }
      ]
    },
    "studentB": {
      "student": { "studentId": 2, "fullName": "Diya Patel" },
      "overallAverage": 71.33,
      "subjectAverages": [
        { "subjectName": "Mathematics", "average": 77.0 },
        { "subjectName": "Science",     "average": 60.0 }
      ]
    },
    "winner": { "studentId": 1, "fullName": "Aarav Sharma", "margin": 16.34 }
  }
}
```

6.4 Student Score Trend Across Exams (by Subject)

GET	`/api/reports/students/{studentId}/trend`	Track a student's score progression over multiple exam events for a given subject

Path Parameters:

Parameter	Type	Description
studentId	`Integer`	The student to track

Query Parameters:

Parameter	Type	Description
subjectId	`Integer`	(Optional) Scope trend to a specific subject

Response — 200 OK

```
{
  "status": "success",
  "data": {
    "student": { "studentId": 1, "fullName": "Aarav Sharma" },
    "subject": { "subjectId": 101, "subjectName": "Mathematics" },
    "trend": [
      { "examId": 1, "examName": "Half-Yearly 2024",  "examDate": "2024-09-15", "score":
85 },
      { "examId": 2, "examName": "Final Annual 2024", "examDate": "2024-03-20", "score":
88 }
    ],
    "trendDirection": "Improving",
    "changePercent": 3.53
  }
}
```

7. Standard Error Response Format

All endpoints return a consistent error envelope on failure:

Error Response Envelope

```
{
  "status": "error",
  "errorCode": "STUDENT_NOT_FOUND",
  "message": "Student with ID 99 does not exist.",
  "timestamp": "2025-03-25T10:30:00Z",
  "path": "/api/students/99"
}
```

Global HTTP Status Codes:

HTTP Status	Meaning
`200 OK`	Request succeeded
`201 Created`	Resource created successfully
`400 Bad Request`	Invalid request payload or missing required fields
`404 Not Found`	Requested resource does not exist
`409 Conflict`	Duplicate entry or referential integrity violation
`422 Unprocessable Entity`	Validation errors on request data (e.g., score > 100)
`500 Internal Server Error`	Unexpected server-side error

8. Complete Endpoint Reference

Method	Endpoint	Description
GET	/api/students	List all students
GET	/api/students/{studentId}	Get student by ID
POST	/api/students	Create new student
PUT	/api/students/{studentId}	Update student
DELETE	/api/students/{studentId}	Delete student
GET	/api/subjects	List all subjects
GET	/api/subjects/{subjectId}	Get subject by ID
POST	/api/subjects	Create new subject
PUT	/api/subjects/{subjectId}	Update subject
DELETE	/api/subjects/{subjectId}	Delete subject
GET	/api/exams	List all exams
GET	/api/exams/{examId}	Get exam by ID
POST	/api/exams	Schedule new exam
PUT	/api/exams/{examId}	Update exam
DELETE	/api/exams/{examId}	Delete exam
GET	/api/marks	List marks (filterable)
GET	/api/marks/{markId}	Get mark by ID
POST	/api/marks	Record new mark
PUT	/api/marks/{markId}	Update mark (correction)
DELETE	/api/marks/{markId}	Delete mark
GET	/api/performance/students/{studentId}	Student full performance
GET	/api/performance/students/{studentId}/exams/{examId}	Student in specific exam
GET	/api/performance/subjects/averages	Subject-wise averages
GET	/api/performance/top-performers	Global top performers
GET	/api/performance/exams/{examId}/pass-fail	Pass/Fail report for exam
GET	/api/reports/exams/{examId}/leaderboard	Exam leaderboard
GET	/api/reports/exams/summary	Exams total score summary
GET	/api/reports/students/compare	Compare two students
GET	/api/reports/students/{studentId}/trend	Student score trend

Total Endpoints: 29 | Base URL: http://localhost:8080 | Format: JSON